南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

计算机组成原理
COMPUTER ORGANIZATION

本课件内容参考自David A. Patterson与John L. Hennessy的教材《Computer Organization and Design: The Hardware/Software Interface》官方配套课件。

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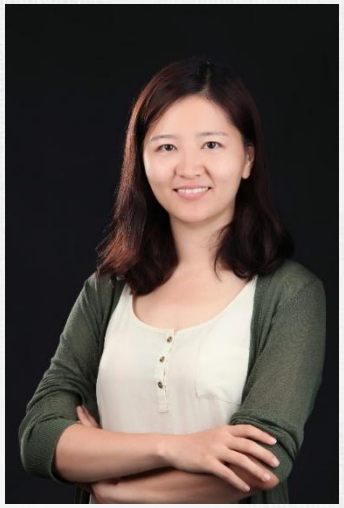
ZHANG, Jin (张进)



- 2000-2004，B.E. in EE, Tsinghua University 清华大学电子系学士(优秀毕业生)
- 2004-2006，M.E. in EE, Tsinghua University 清华大学电子系硕士
- 2006-2009，Ph.D. in CS, HKUST 香港科技大学计算机系博士
- 2009-2014，Research Assistant Prof., HKUST 香港科技大学计算机系研究助理教授
- 2014-2021，Assist. Prof., SUSTech 南方科技大学计算机系助理教授
- 2021-， Tenured Associate Prof., SUSTech 南方科技大学计算机系长聘副教授（兼任副系主任，致仁书院学术副院长）

Research Interest: Intelligent Sensing, Edge Computing, Smart Healthcare, Wearable Computing, Blockchain (智能感知，边缘智能，医疗物联网，可穿戴计算，区块链)

- 120 publications in Top Journals and Conferences, 48000+ Citations, H-index: 31 在TWC、TMC、TPDS、TDSC、Mobicom、CHI、Ubicomp、ICDCS等国际顶级期刊会议发表论文120余篇, 被引48000余次，H指数31
- Secured more than 20 fundings and projects in recent years 近五年承担和参与国家、广东省和深圳市科研项目20余项，包括：国家自然科学基金（青年，面上），科技部重点研发计划，省重点实验室，省创新团队，市工程实验室等
- Many awards in teaching 曾获优秀导师奖，优秀教学奖，青年教师教学大赛一等奖，最受毕业生欢迎任课教师奖，深圳市教工委优秀共产党员等

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Computer Organization

Introduction

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Computer Organization in CS Curriculum System

Systems Thinking 系统思维 (how to design a system)	Computational Thinking 计算思维 （how to solve a problem)
Digital Logic 数字逻辑	Programing 程序设计
Comp. Org. 计算机组成	Discrete Math. 离散数学
Operating System 操作系统	Data Structure 数据结构
Database 数据库	Algorithm Design 算法设计
Comp. Networks 计算机网络	Art. Intelligence人工智能
Compiling 编译原理	Software Eng. 软件工程
Parallel Computing 并行计算	

Computer Organization in CS Curriculum System

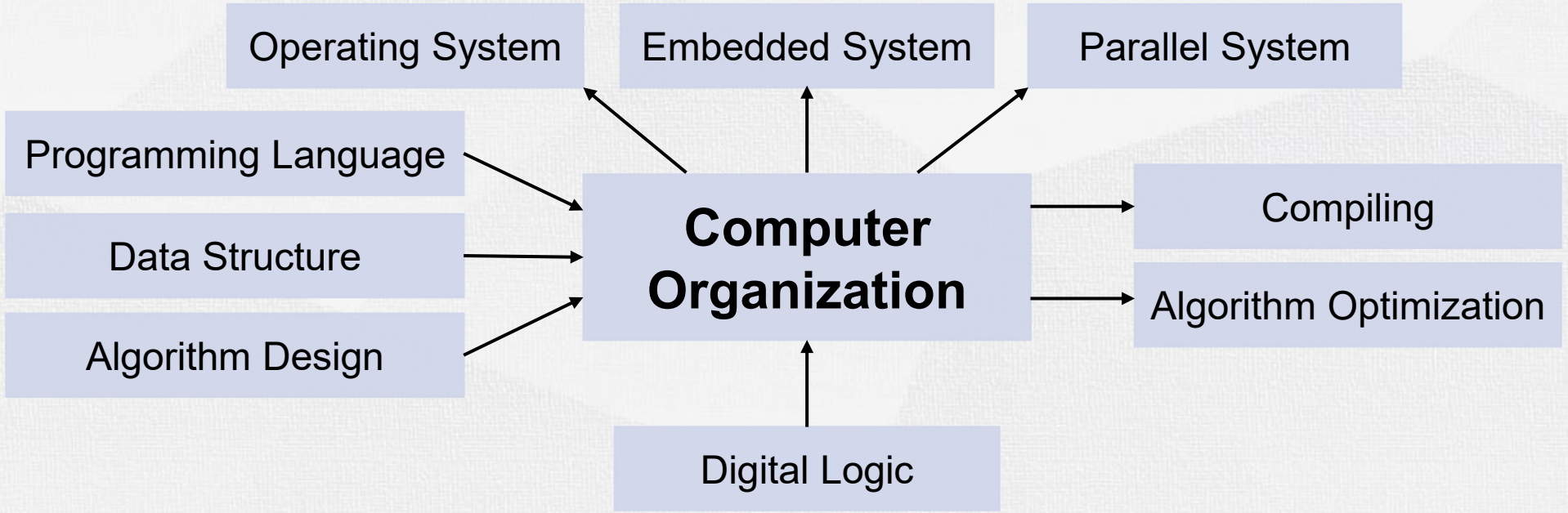
System Thinking 系统思维 (how to design a system)	Computational Thinking 计算思维 （how to solve a problem)	Other Applications 计算机应用
Digital Logic 数字逻辑	Programing 程序设计	Machine Learning机器学习
Comp. Org. 计算机组成	Discrete Math. 离散数学	Deep Learning 深度学习
Operating System 操作系统	Data Structure 数据结构	Comp. Vision 计算机视觉
Database 数据库	Algorithm Design 算法设计	Comp. Security计算机安全
Comp. Networks 计算机网络	Art. Intelligence人工智能	Internet of Things物联网
Compiling 编译原理	Software Eng. 软件工程	Robotics智能机器人
Parallel Computing 并行计算		Cryptography密码学

Computer Organization in CS

- For system thinking
  - Learn how to design a system in Computer Organization course, because computer is an example of a system
  - Computing organization is the hardware foundation of the software systems on above, e.g., operating system, parallel system
- For computational thinking
  - Learn how to solve a problem using a computer, learn how the program is executed in the computer in Computer Organization course
  - helps to solve a problem faster and better, helps to optimize the program

Importance of Computer Organization

- a fundamental and indispensable core course in the computer science curriculum.
- establish a complete and systematic view of computer
- builds a critical bridge between other core courses



Why to Learn Computer Organization?

- Embarrassing if you are a student in CS and can't make sense of the following terms: DRAM, SRAM, pipelining, cache hierarchies, I/O, virtual memory, …
- Embarrassing if you are a student in CS and can't decide which processor to buy: 3 GHz P4 or 2.5 GHz Athlon (this course helps us reason about performance/power), …
- First step for chip designers, compiler/OS writers
- Knowledge of the hardware will help you write better programs

Must a Programmer Care about Hardware?

- Must know how to reason about program performance and energy
- CPU Performance: if we understand how CPU process data, we can enhance the computation efficiency
- Memory management: if we understand how/where data is placed, we can help ensure that relevant data is nearby
- Thread management: if we understand how threads interact, we can write smarter multi-threaded programs
- I/O management

Prerequisites

- Binary numbers
- Read and write basic C/Java programs
- Understand the steps in compiling and executing a program
- Digital Circuit, Logic design:
  - Logical equations, schematic diagrams
  - Combinational vs. sequential logic
  - Finite state machines (FSMs)

What You will Learn

- Major content
  - Basic parts of a computer (processor, memory, disk, etc.)
  - Principles of computer architecture: CPU datapath and control unit design
  - Assembly language programming in RISC-V
  - Memory hierarchies and design
  - I/O organization and design
- Course goals
  - To learn the organizational structures that determine the capabilities and performance of computer systems
  - To understand the interactions between the computer's architecture and its software
  - To understand cost performance trade-offs

Key Topics

- Introduction (Chapter 1)
  - Basic terms
  - Moore's Law, power wall
  - Core ideas in computer architecture
- Processors (Chapter 2-4)
  - Assembly language (Chapter 2)
  - Computer arithmetic (Chapter 3)
  - Pipelining (Chapter 4)
- Memory (Chapter 5)
- Parallel Processors (Chapter 6)

What You will Learn

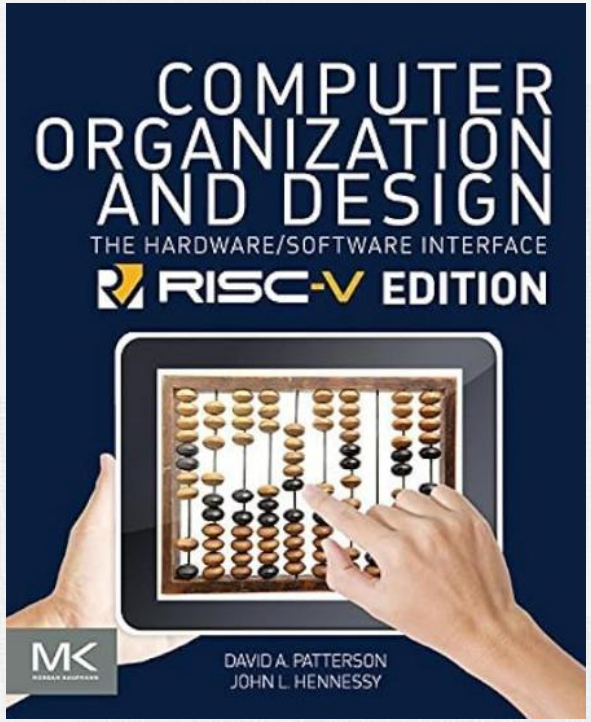
- How programs are translated into the machine language
  - And how the hardware executes them
- The hardware/software interface
- What determines program performance
  - And how it can be improved
- How hardware designers improve performance
- What is parallel processing

Understanding Performance

- Algorithm
  - Determines number of operations executed
- Programming language, compiler, architecture
  - Determine number of machine instructions executed per operation
- Processor and memory system
  - Determine how fast instructions are executed
- I/O system (including OS)
  - Determines how fast I/O operations are executed

Textbooks

- Textbook:
  - Computer Organization & Design, the Hardware/Software Interface, RISC-V edition. D. A. Patterson and J. L. Hennessy
    - The Textbook uses RV64, in class we learn **RV32**



Turing award 2017

David A. Patterson
Professor of UC Berkeley
Distinguished Engineer at Google

John L. Hennessy
President of Stanford University
Chairman of Alphabet

Recommended Readings

- 《编码——隐匿在计算机软硬件背后》
- 《黄仁勋：英伟达之芯》、《埃隆·马斯克传》、《史蒂夫·乔布斯传》、《爱因斯坦传》、《本杰明·富兰克林传》、《创新者：一群技术狂人和鬼才程序员如何改变世界》
- 《浪潮之巅 》《数学之美》吴军
- 《沸腾十五年》《沸腾新十年》《激荡四十年》吴晓波
- 《黑客与画家》硅谷创业之父Paul Graham
- 《人月神话》《信息简史》
- 《人类简史》《今日简史》《智人之上》尤瓦尔·赫拉利

Changes of the World's Largest Companies

- General Electric (1990) → Microsoft (1998) → Apple (2012) → Nvidia (2024)
- Industrial → Software in PC era → Hardware and software in Mobile era → infrastructure in AI era
- Nvidia's AI infrastructure: GPU+CUDA
- Meanwhile, there are more and more recent IPO or unicorn companies working on AI infrastructure: 寒武纪、壁仞、沐曦、地平线
- Computer organization will tell you the foundation of how to design a GPU/NPU/TPU…

Grading Policy

- 30% Mid-term examination
  - **Midterm: 2026/04/25 10am-12am** (Lecture 1-8)
- 30% Final examination
- 5% Lecture Attendance
- 5% Homework
- 5% Lab Attendance and practices
- 10% Lab assignments – RISC-V & Verilog Programming
- 15% Lab Project
  - In groups of 2~3. Please team up as soon as possible

Honor Policy

- All course work should be completed entirely on your own. You are encouraged to discuss general concepts and ideas in homework or lab assignments.
- Students who commit an act of academic dishonesty may receive a zero on the assignment (first conduct) or in the course (multiple conducts).
- Unless otherwise noted, exams and individual assignments will be pledged that you have neither given nor received unauthorized help.
- If you have questions on what is allowable, ask!

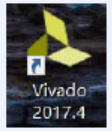
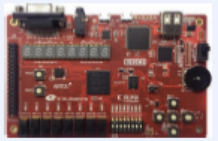
Course Schedule

WEEK	LECTURE	LECTURE TOPIC	LAB CONTENT
1	Lecture #1	Introductions	Environment Setup
2	Lecture #2	RISC-V ISAs: Basics	Assembly Practice 1
3	Lecture #3	RISC-V ISAs: Procedure Call	Assembly Practice 2
4	Lecture #4	RISC-V ISAs: Addressing	Assembly Practice 3
5	Lecture #5	Performance	Assembly Practice 4
6	Lecture #6	Arithmetic	Assembly Practice 5
7	Break	/	Assembly Practice 6
8	Lecture #7	Floating Point Arithmetic	Hardware vs Software CPU
9	**Lecture #8**	**The Processor** **(Midterm 2026/04/25 10am-12am)**	**CPU Design 1**
10	Lecture #9	The Pipeline	CPU Design 2
11	Break	/	CPU Design 3
12	Lecture #10	Instruction-Level Parallelism	CPU Design 4
13	Lecture #11	Memory Hierarchy1	CPU Design 5
14	Lecture #12	Memory Hierarchy2	CPU Design 6
15	Lecture #13	Parallel Processors	Project Inspection

Lab ToolChain

- Labs are a key portion of the class
- Highly recommended to find your partner as soon as possible

Tools used in our Labs

Task	Tool kits
Learn and practice **RISC-V**(a type of Assemblly language)	➢ **Rars (rars_27a7c1f)**
Design and implement an **CPU**	➢ **Vivado**  ➢ **FPGA based Development Board EGO1** 
Test the **CPU** with **program(s)**, both of which are based on **RISC-V**	➢ **Assembler (Rars)** ➢ **Uart Tools** ➢ **Vivado** ➢ **FPGA based Development Board(EGO1)**

Tips for Attending the Lecture

- To get the best use of lecture
  - be interactive
  - ask whenever you have question, interrupt whenever you want
  - ask immediately after the class if you are shy
  - give me suggestions and feedback frequently
- Get the main idea in class, read the details after class

The Computing Evolution and Computer Hardware

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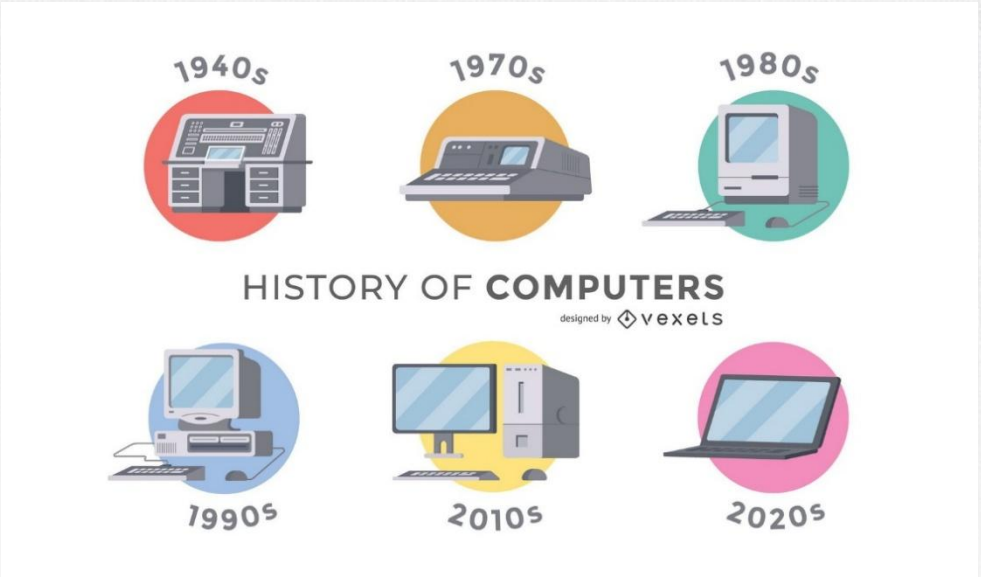
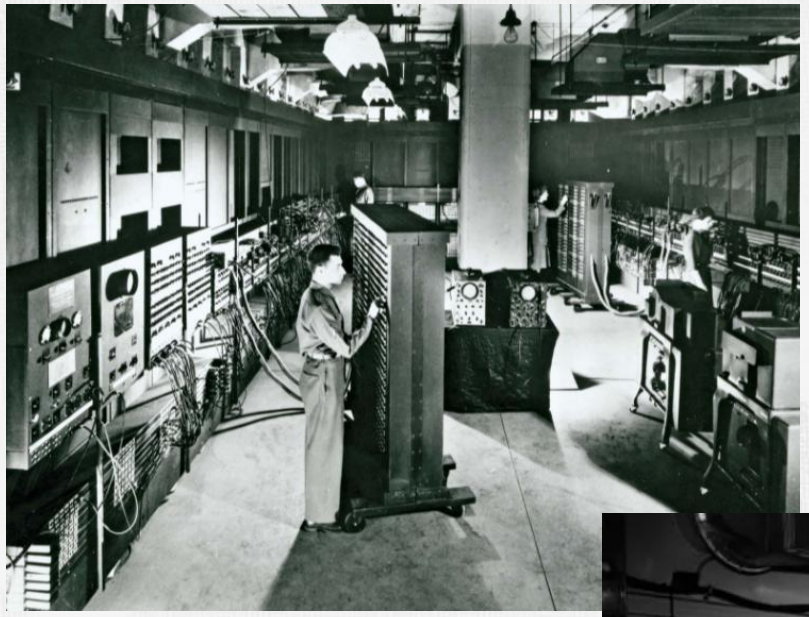
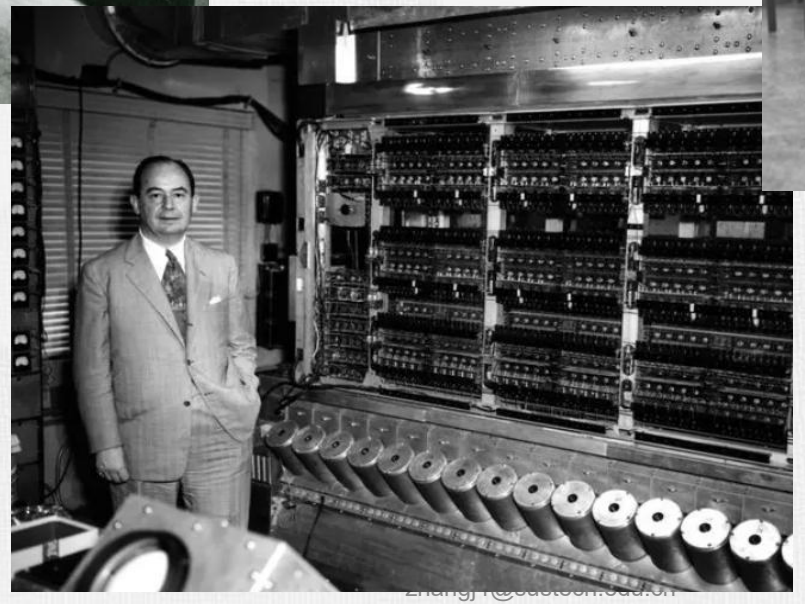
The Evolution of Computers

- First Generation (1940s - 1950s)
  - Vacuum Tubes
- Second Generation (1950s - 1960s)
  - Transistors
- Third Generation (1960s - 1970s)
  - Integrated Circuits
- Fourth/Now Generation (1970s - Present)
  - Microprocessors
  - Artificial Intelligence



Old School Computers





New School Computers

From the Small…

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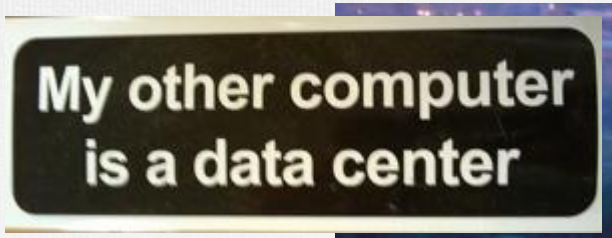
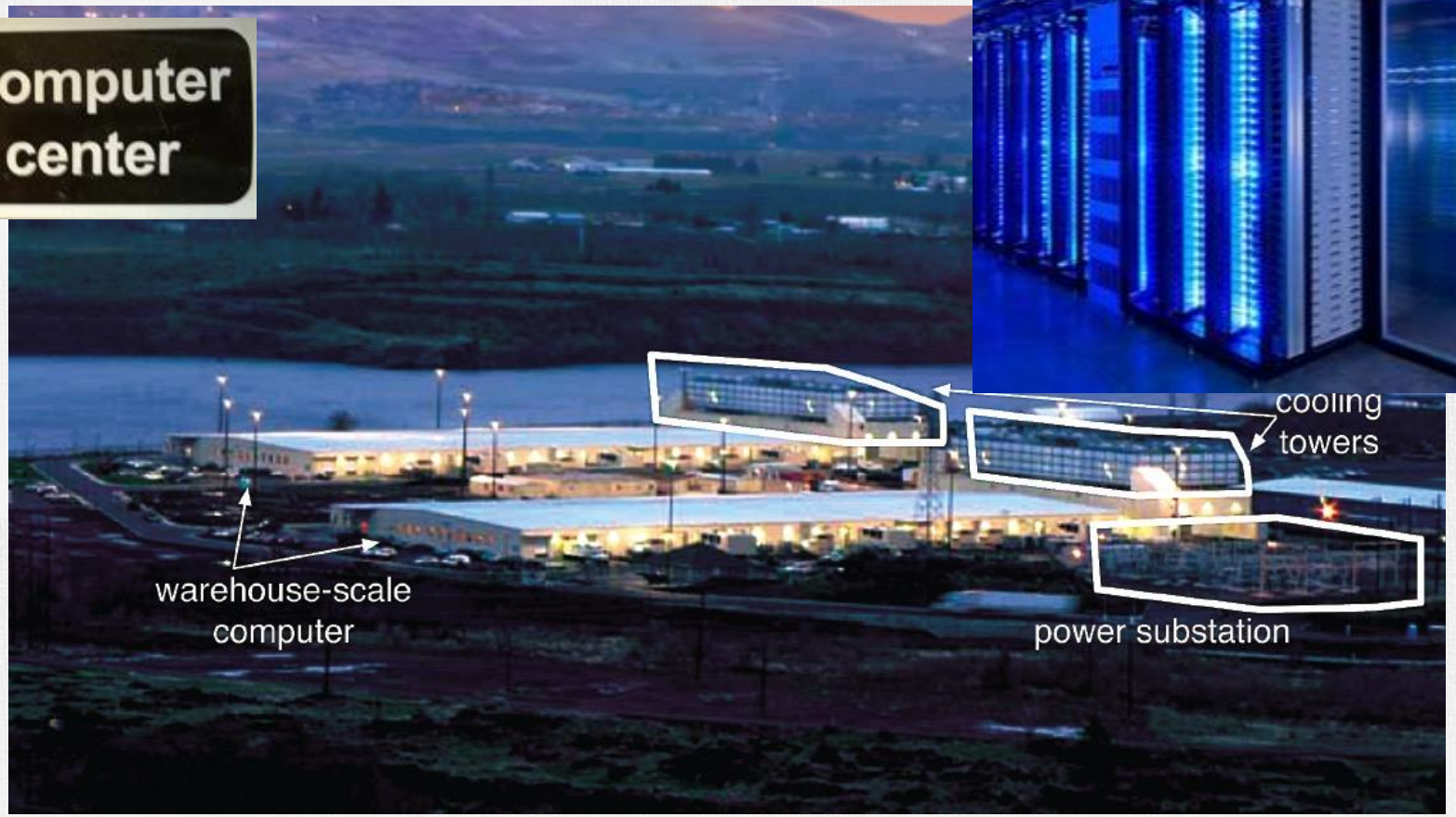
New School Computers

To the very small...

New School Computers

To the big…





The Computer Revolution

- Progress in computer technology
  - Underpinned by Moore's Law
- Makes novel applications feasible
  - Computers in automobiles
  - Cell phones
  - Human genome project
  - World Wide Web
  - Search Engines
- Computers are pervasive

Classes of Computers

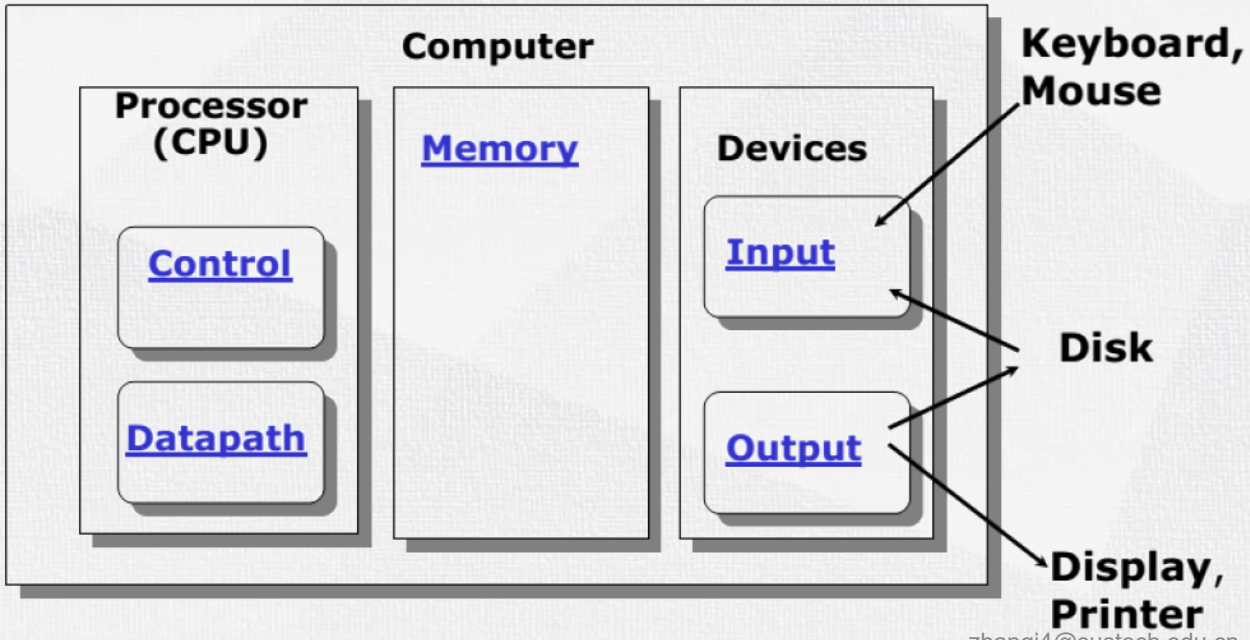
- Personal computers
  - General purpose, variety of software
  - Subject to cost/performance tradeoff
- Server computers
  - Network based
  - High capacity, performance, reliability
  - Range from small servers to building sized
- Supercomputers
  - High-end scientific and engineering calculations
  - Highest capability but represent a small fraction of the overall computer market
  - https://www.top500.org/lists/top500
- Embedded computers
  - Hidden as components of systems
  - Stringent power/performance/cost constraints

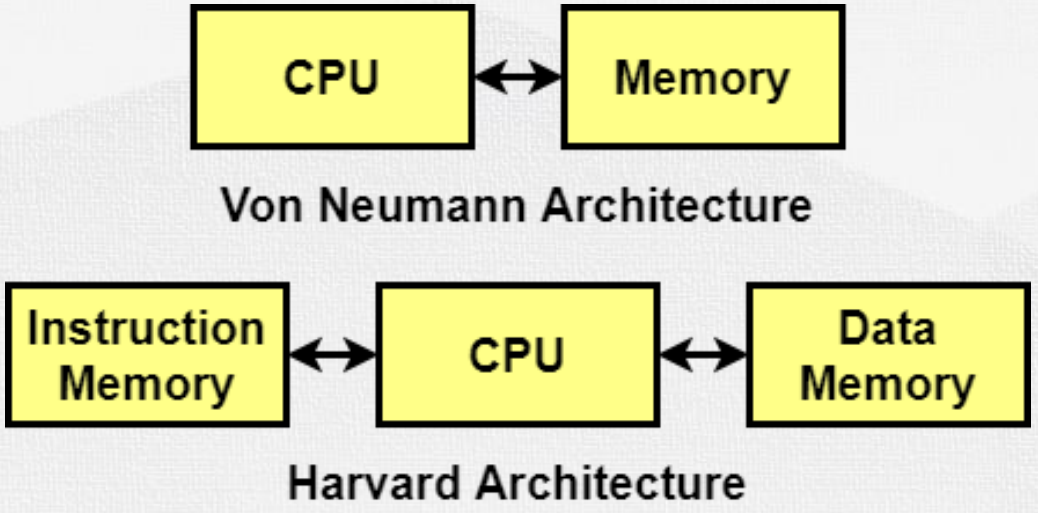
Classes of Computers in The PostPC Era

- Personal Mobile Device (PMD)
  - Battery operated
  - Connects to the Internet
  - Hundreds of dollars
  - Internet of Things
- Cloud computing
  - Warehouse Scale Computers (WSC)
  - Software as a Service (SaaS)
  - Data Centers
- New Architecture for AI Era
  - More Suitable for deep learning
  - GPU, TPU, NPU
  - AI Chips

Components of a Computer

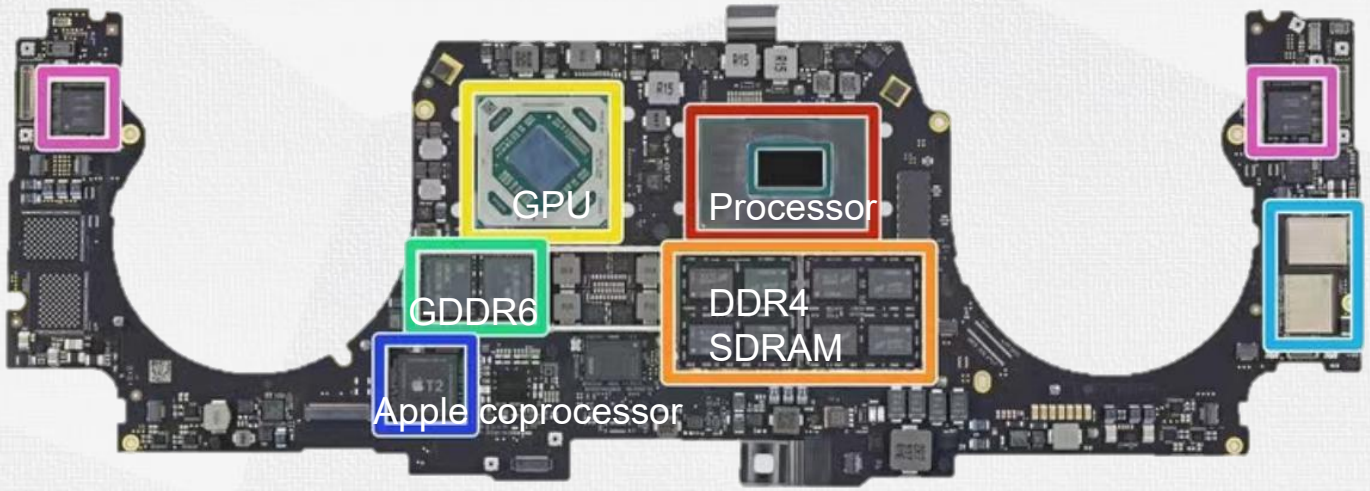
- Same components for all kinds of computer:
  - Input, Output, Memory, Processor (Control, Datapath/ALU)
- Von Neumann Architecture
- Harvard Architecture



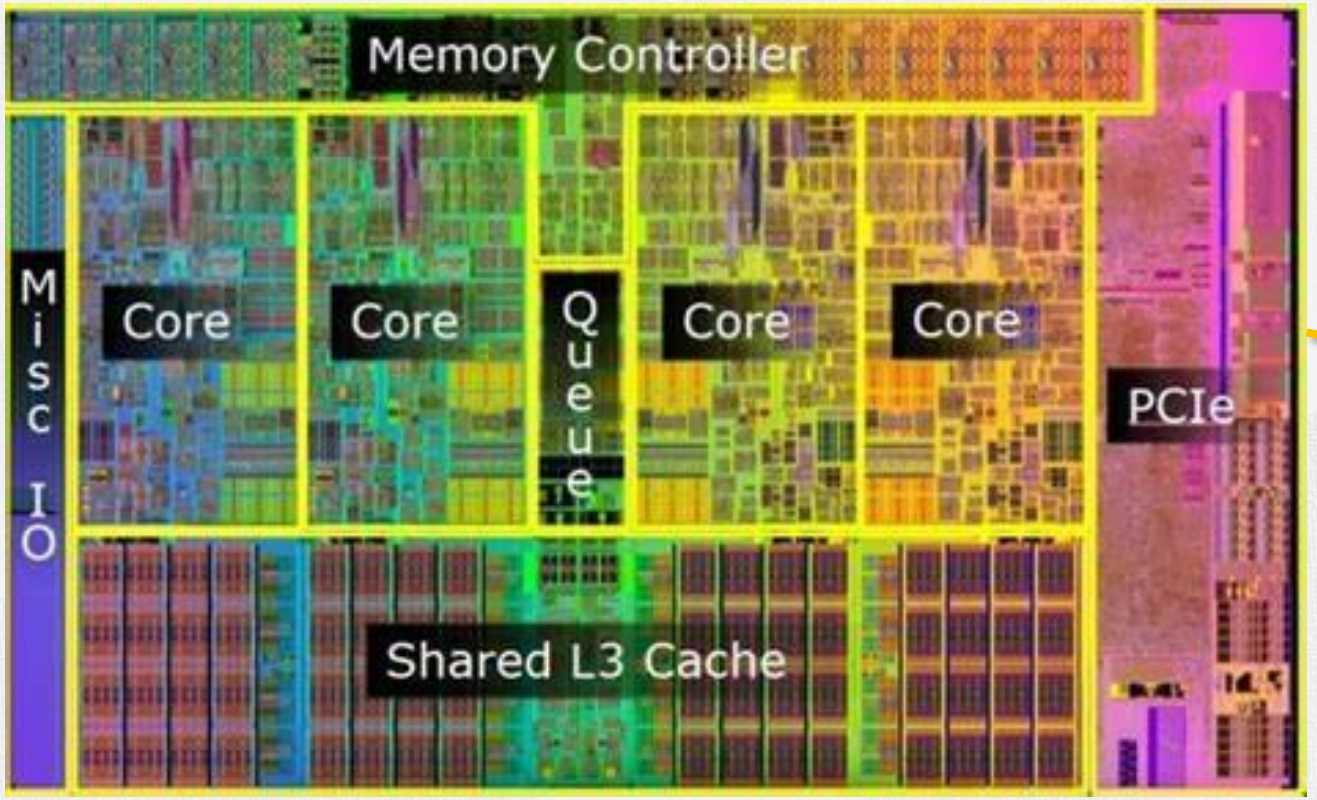


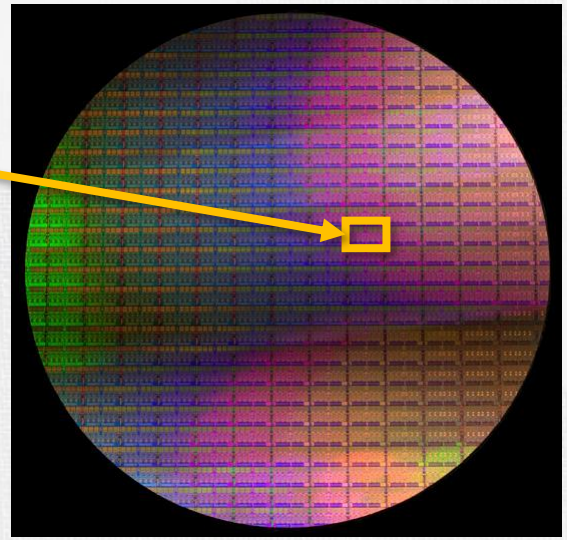
Teardown of MacBook

- 16" LED-backlit IPS Retina display
- Keyboard and Touch Bar
- 2.6 GHz 6-core Intel Core i7
- 16 GB of 2666 MHz DDR4 SDRAM
- 512 GB SSD
- 100 Watt-hour battery
- Speaker and microphone



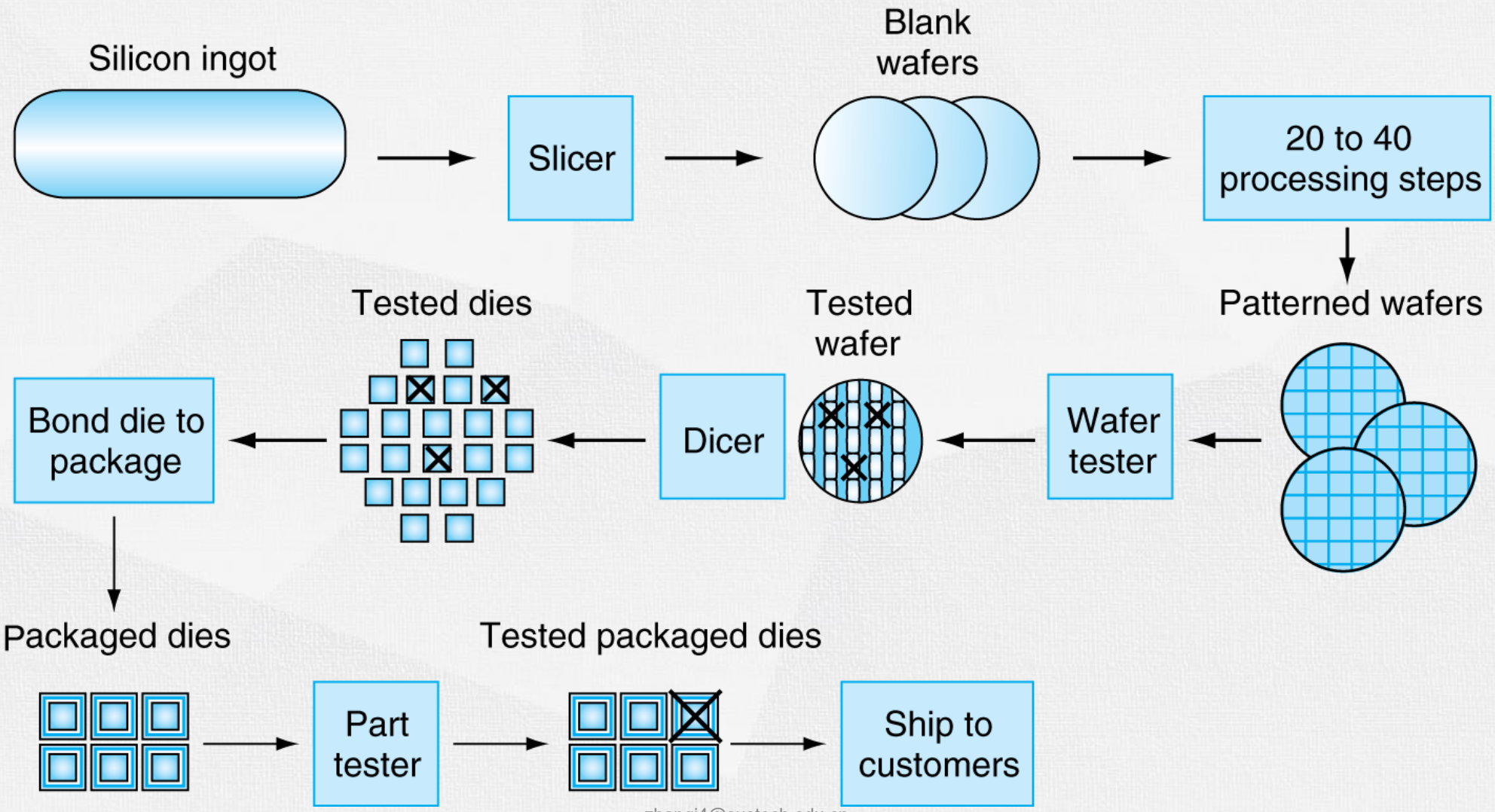
Inside the Processors



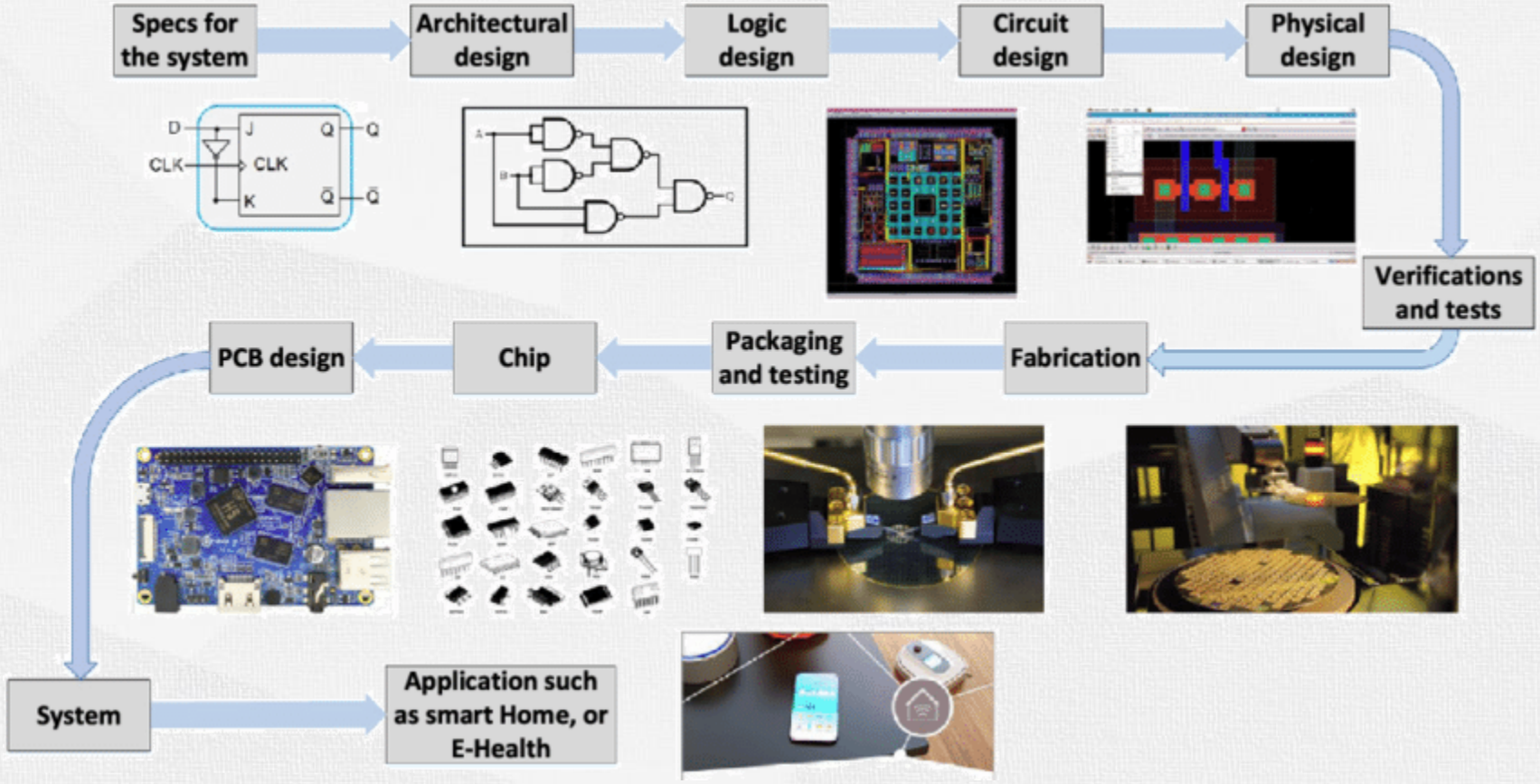


Intel's Core i7 wafer and Die map

IC Fabrication

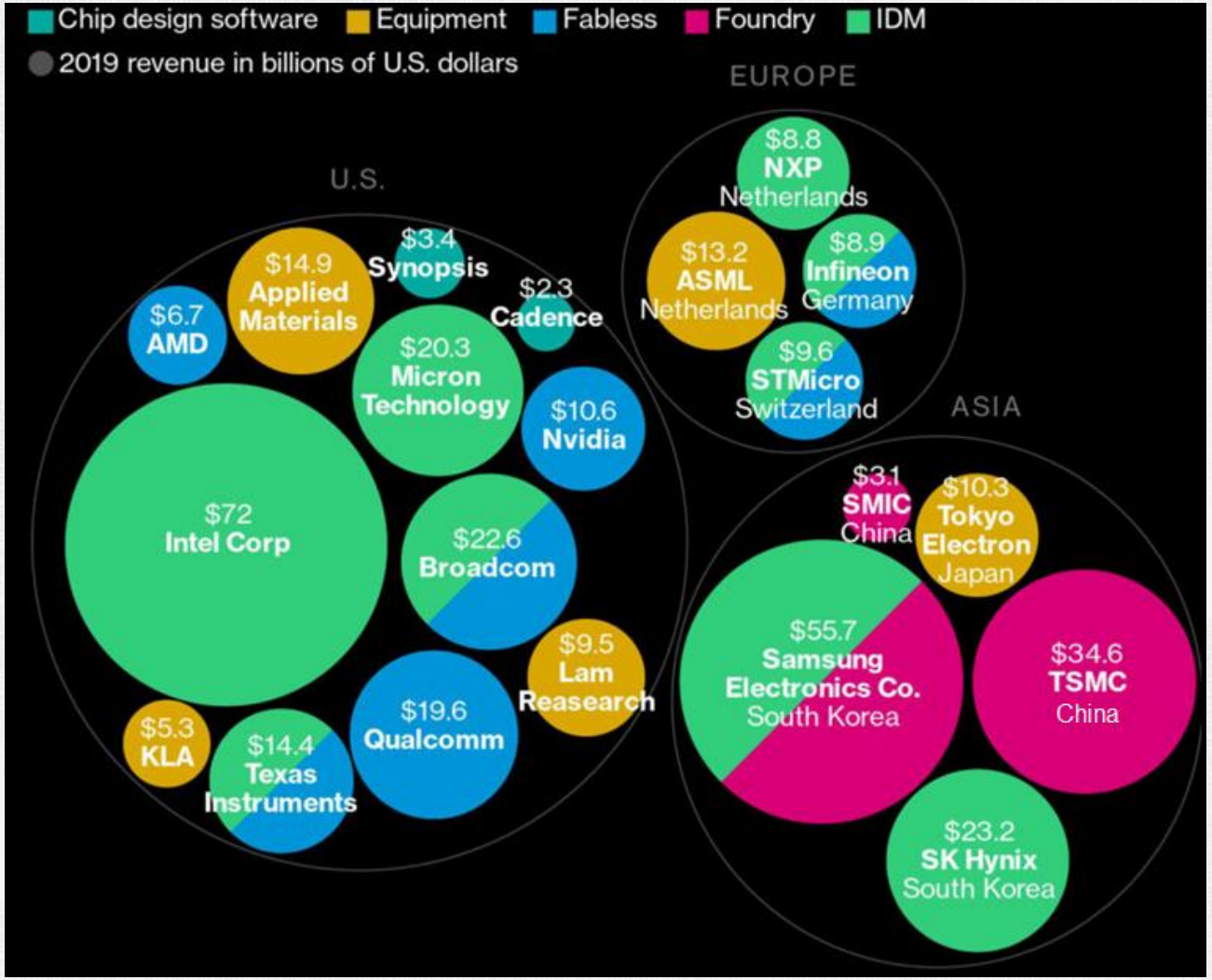


Semiconductor Manufacturing Process Chain



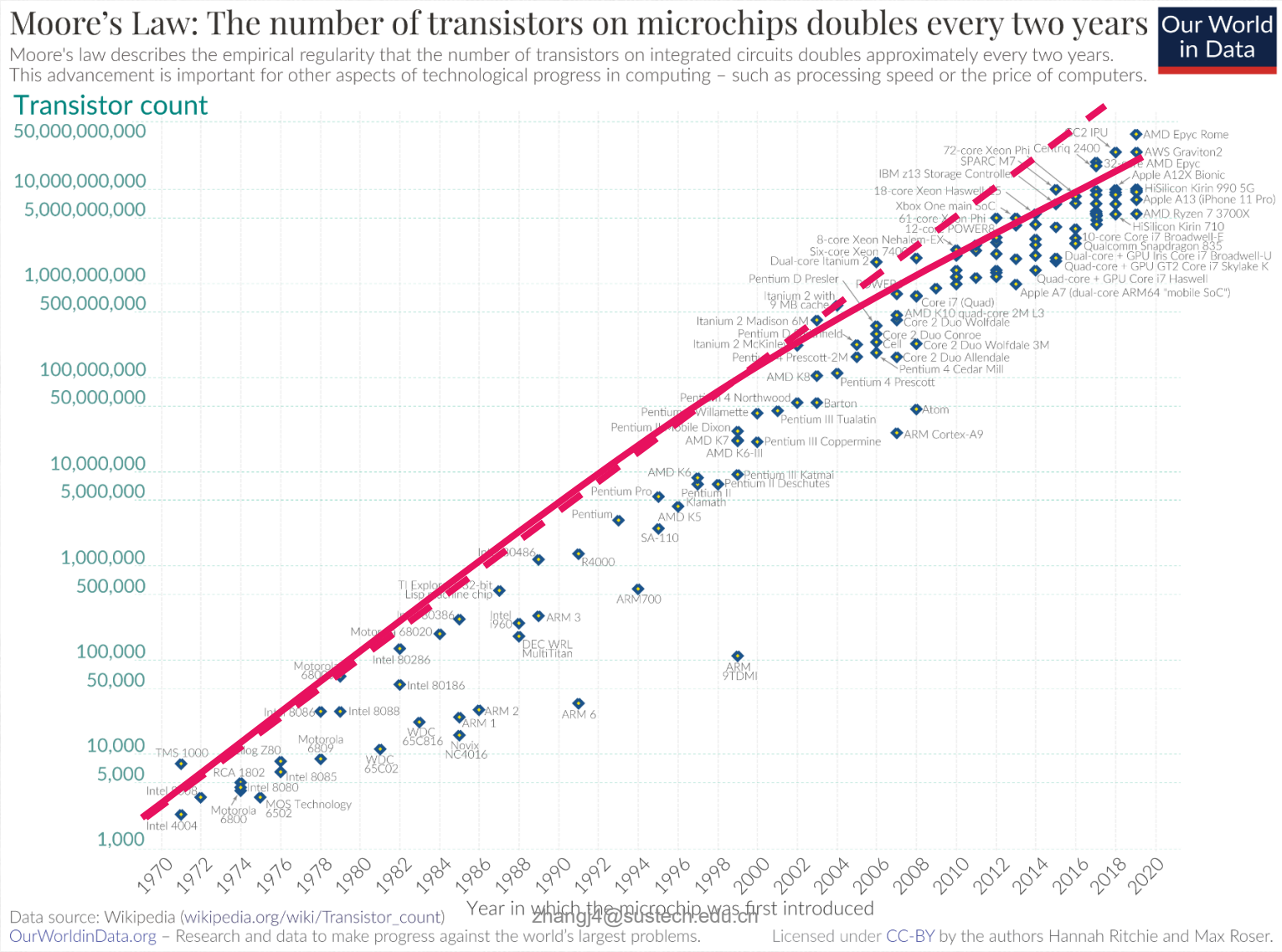
Chip Industry

- Key players in chip industry
  - Intel
  - AMD
  - Qualcomm
  - Samsung
  - TSMC
  - Broadcom
  - Nvidia
  - ASML
  - …



IDM: Integrated Design and Manufacture

Moore’s Law


Moore’s Law: The number of transistors on microchips doubles every two years
Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years.
This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.
Our World in Data
Transistor count
50,000,000,000
10,000,000,000
5,000,000,000
1,000,000,000
500,000,000
100,000,000
50,000,000
10,000,000
5,000,000
1,000,000
500,000
100,000
50,000
10,000
5,000
1,000
1970
1972
1974
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1980
1982
1984
1986
1988
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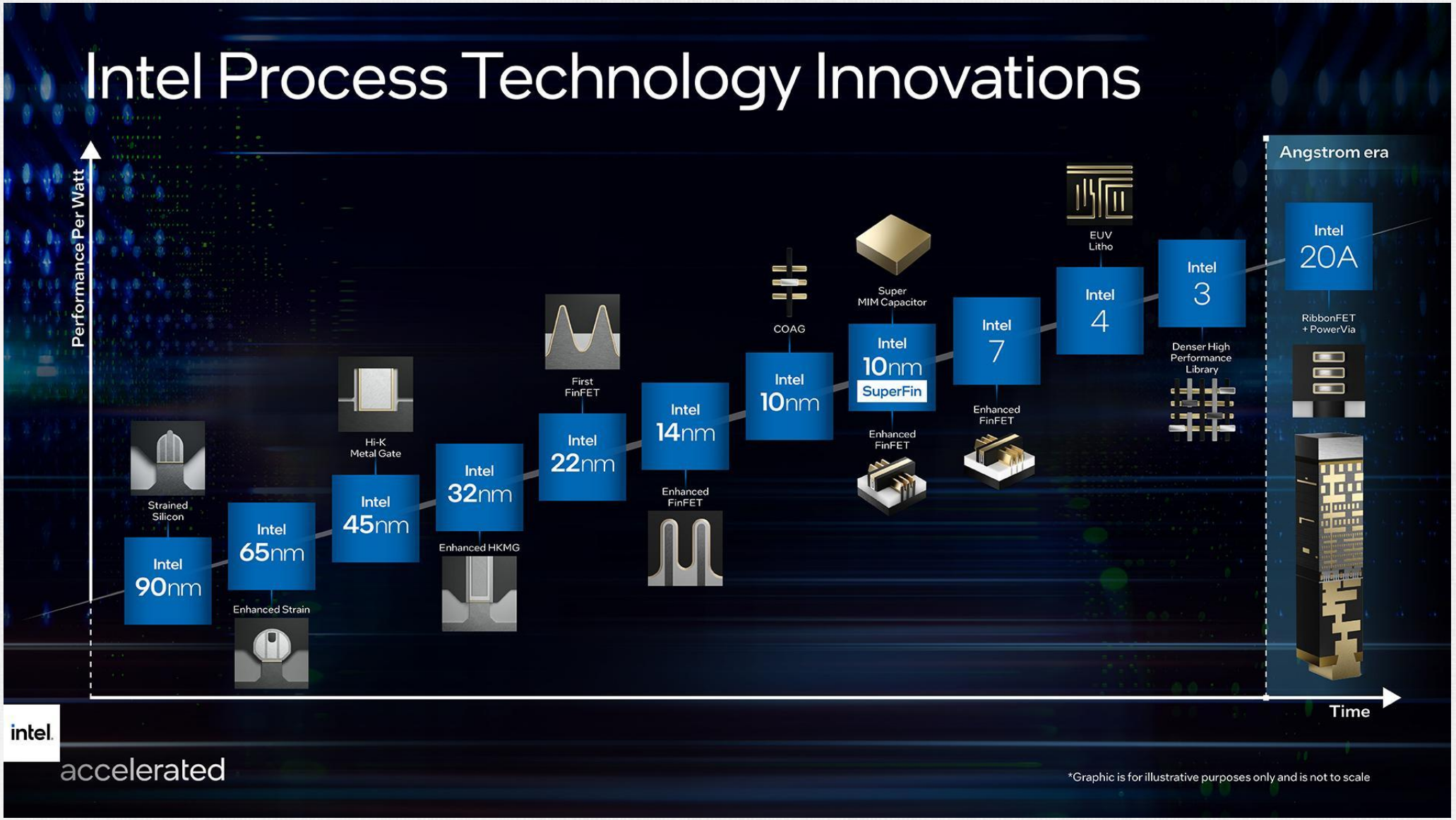
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
Year in which the microchip was first introduced
Intel 4004
TMS 1000
RCA 1802
Intel 8080
Motorola 6800
MOS Technology 6502
Intel 8085
Motorola 6809
Intel 8088
Intel 80186
Intel 80286
Motorola 68020
WDC 65C02
WDC 65C816
Novix NC4016
ARM 1
ARM 2
ARM 3
ARM 6
ARM 9TDMI
DEC WRL MultiTitan
Intel i960
R4000
ARM700
Pentium
SA-110
AMD K5
Pentium Pro
Pentium II Klamath
Pentium II Deschutes
Pentium III Katmai
AMD K6
AMD K6-III
AMD K7
Pentium III Coppermine
Pentium III Tualatin
Pentium 4 Northwood
Barton
AMD K8
Pentium 4 Prescott
Pentium 4 Cedar Mill
Core 2 Duo Allendale
Core 2 Duo Wolfdale 3M
Core 2 Duo Conroe
Core 2 Duo Wolfdale
AMD K10 quad-core 2M L3
Core i7 (Quad)
Quad-core + GPU Core i7 Haswell
Apple A7 (dual-core ARM64 "mobile SoC")
Quad-core + GPU GT2 Core i7 Skylake K
Dual-core + GPU Iris Core i7 Broadwell-U
Qualcomm Snapdragon 835
10-core Core i7 Broadwell-E
HiSilicon Kirin 710
AMD Ryzen 7 3700X
Apple A13 (iPhone 11 Pro)
HiSilicon Kirin 990 5G
Apple A12X Bionic
AWS Graviton2
AMD Epyc Rome
Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)
OurWorldinData.org – Research and data to make progress against the world's largest problems.
Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

Processor Technology Trends



Core Ideas in Computer Architecture and Importance of Abstraction

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Eight Great Ideas

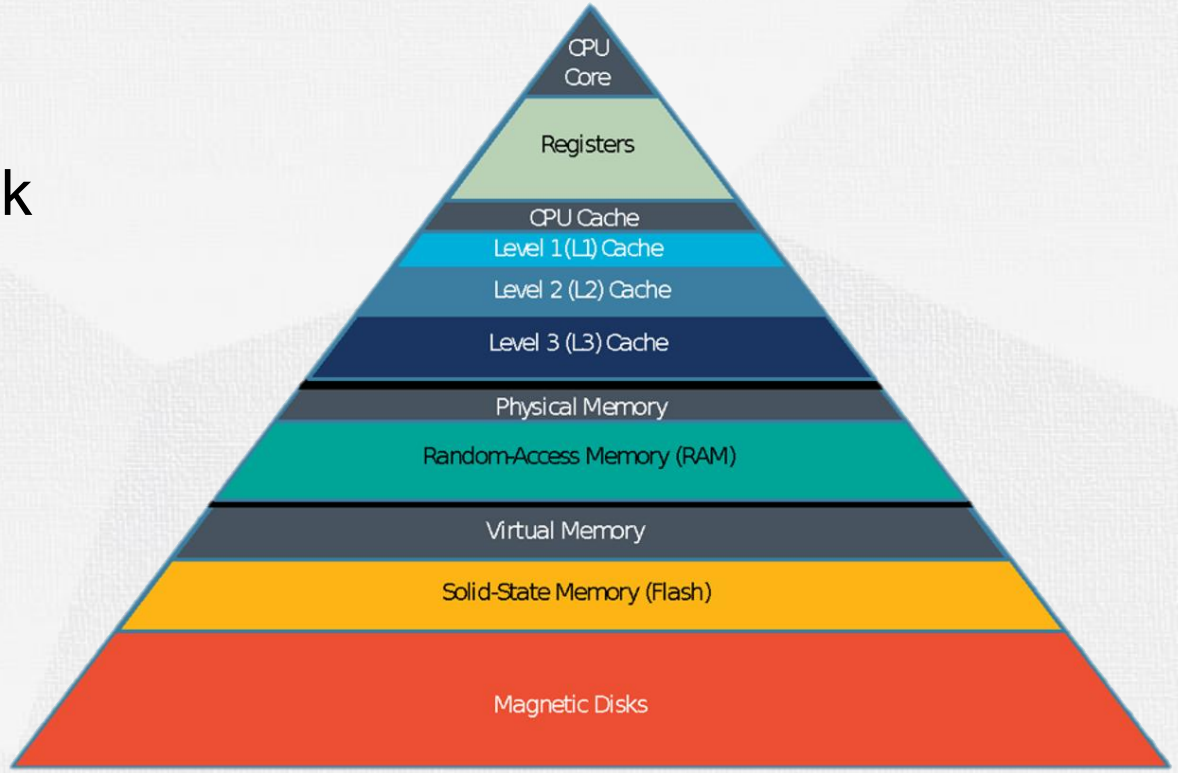
- Design for ***Moore's Law***
- Use ***abstraction*** to simplify design
- Make the ***common case fast***
- Performance *via* ***parallelism***
- Performance *via* ***pipelining***
- Performance *via* ***prediction***
- ***Hierarchy*** of memories
- ***Dependability*** *via* redundancy

Ways to Improve Performance

- Common Case Fast:
  - Enhances performance better than optimizing the rare case
    - The common case is often simpler than rare case and, hence, it is easier to enhance.
- Performance via Parallelism:
  - Running multiple operations in parallel enhances the computer performance
- Performance via Pipelining:
  - A particular pattern of parallelism
  - A set of data processing elements connected in series, so that the output of one element is the input of the next one
- Performance via Prediction
  - It can be faster on average to guess and start working rather than wait

Hierarchy of Memories

- The fastest, smallest, and most expensive memory per bit at the top of the hierarchy;
- The slowest, largest, and cheapest per bit at the bottom.
  - Cache (SRAM)
  - Main memory (DRAM)
  - Secondary Storage: e.g. Hard disk



Dependability via Redundancy

- Any physical device, including a computer, can fail:
  - We make systems dependable by including the redundant components that can take over when a failure occurs and help detect failures.

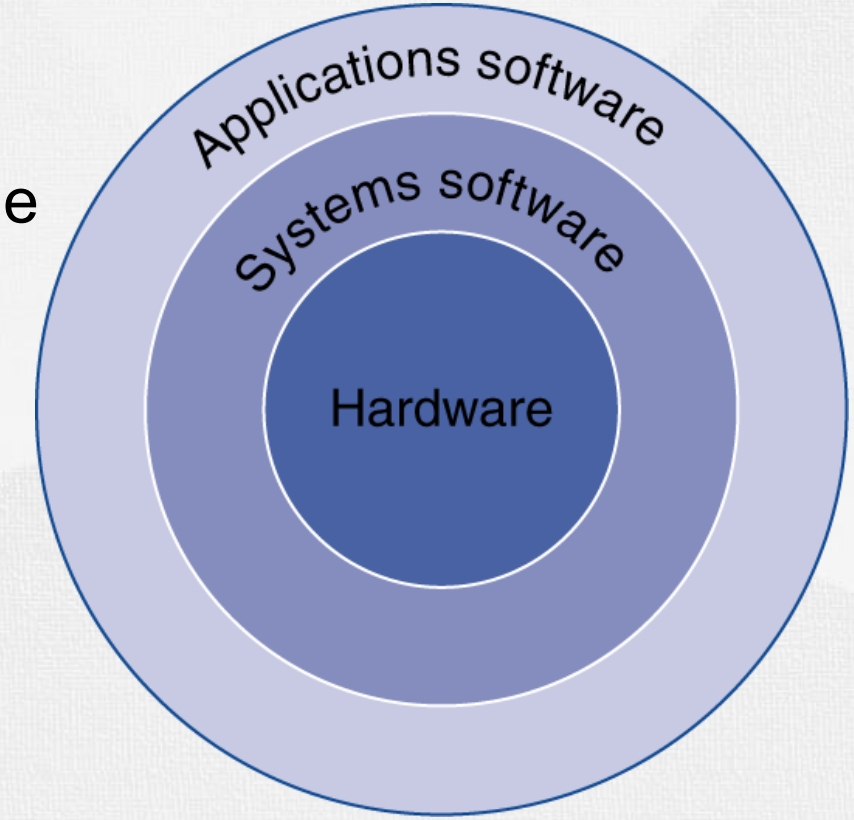
Design Rules

- Design for Moore's Law
  - Computer architects must anticipate where the technology will be when the design finishes rather than design for where it starts

- Abstraction
  - Increases productivity for hardware and software
    - Lower-level details are hidden to offer a simpler model at higher levels

Below Your Program

- Application software
  - Written in high-level language
- System software
  - Compiler: translates HLL code to machine code
  - Operating System: service code
    - Handling input/output
    - Managing memory and storage
    - Scheduling tasks & sharing resources
- Hardware
  - Processor, memory, I/O controllers



Levels of Program Code

- High-level language
  - Level of abstraction closer to problem domain
  - Provides for productivity and portability
- Assembly language
  - Textual representation of instructions
- Hardware representation
  - Binary digits (bits)
  - Encoded instructions and data

High-level language program (in C)

```
swap(int v[], int k)
{int temp;
   temp = v[k];
   v[k] = v[k+1];
   v[k+1] = temp;
}
```

Compiler

Assembly language program (for RISC-V)

```
swap:
      slli x6, x11, 3
      add  x6, x10, x6
      ld   x5, 0(x6)
      ld   x7, 8(x6)
      sd   x7, 0(x6)
      sd   x5, 8(x6)
      jalr x0, 0(x1)
```

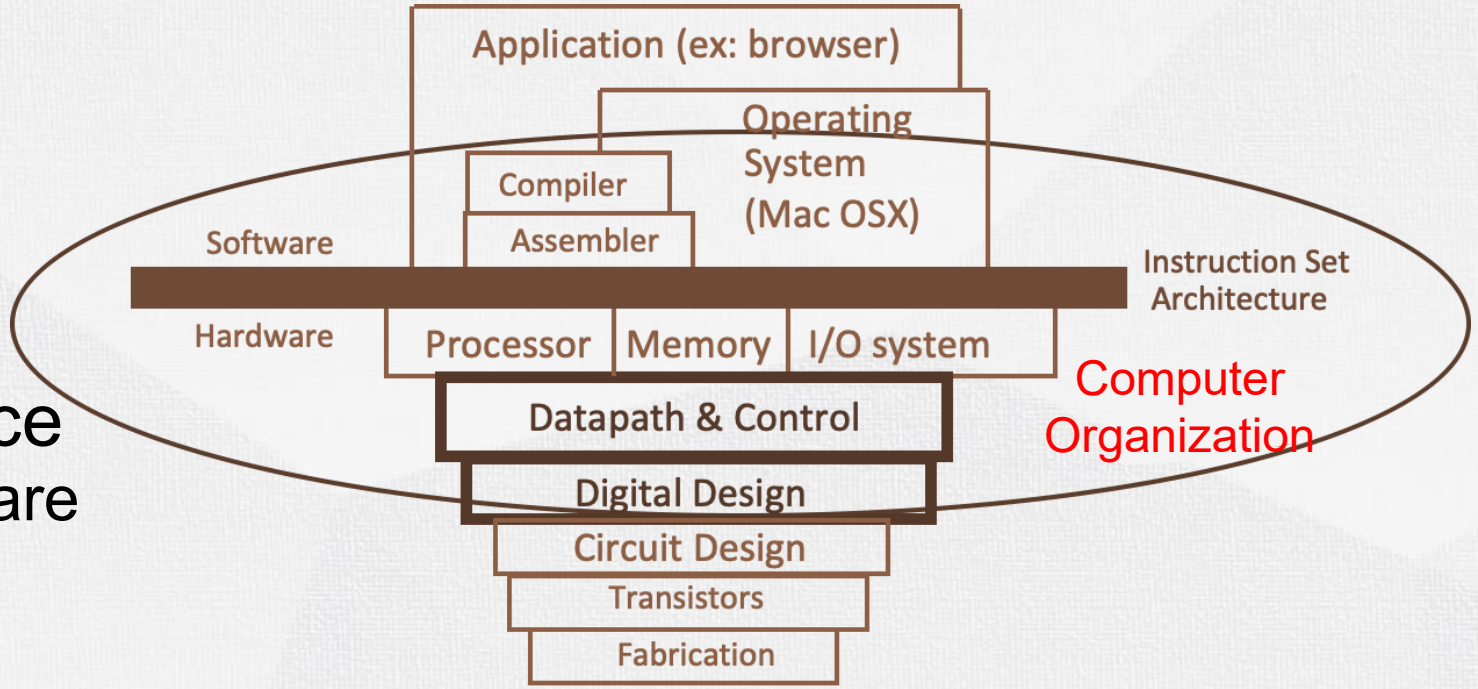
Assembler

Binary machine language program (for RISC-V)

```
00000000001101011001001100010011
00000000011001010000001100110011
00000000000000110011001010000011
00000000100000110011001110000011
00000000011100110011000000100011
00000000010100110011010000100011
00000000000000001000000001100111
```

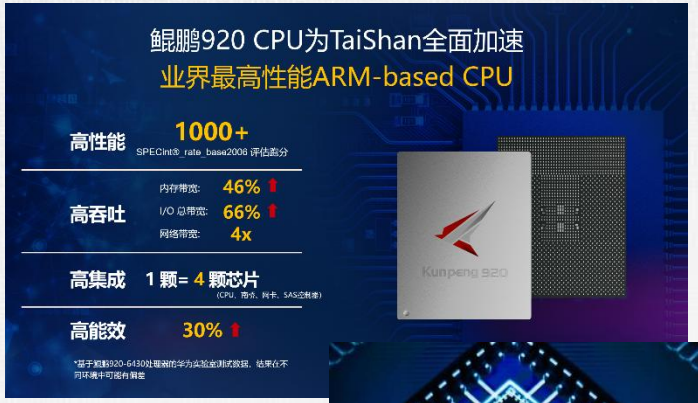
Abstractions

- Abstraction helps us deal with complexity
  - Hide lower-level detail
- **Instruction set architecture (ISA)**
  - The hardware/software interface
- Application binary interface
  - The ISA plus system software interface
- Implementation
  - The details underlying and interface



ISA: RISC vs CISC

	RISC (Reduced Instruction Set Computer)	CISC (Complex Instruction Set Computer)
Example	Instruction: Pick up spoon → Scoop rice → Bring to mouth → Swallow	Instruction: Eat meal
Pros & Cons	• Simple, optimized for common operations; can be pipelined efficiently • Multiple instructions needed for one complex task • Execution cycles: ≤ 1 cycle per instruction	• Single instruction does it all, dense code • Complex instruction, slow execution; pipeline stages often wasted; complex hardware design • Execution cycles: Typically 5-8 cycles per instruction
Types	ARM, RISC-V, MIPS, LoongArch	x86

RISC Architectures & Chip Examples

- ARM
  - **Huawei Kunpeng 920 (server)**
  - Apple M1 (Laptop)
  - Qualcomm Snapdragon (Mobile)
- RISC-V
  - **Alibaba T-Head Xuantie C910 (HPC)**
  - SiFive Intelligence X160 (AI acceleration)
  - ESP32-C6 (IoT)
- MIPS
  - Microchip PIC32 (Industrial control)
- LoongArch
  - **Loongson 3A6000 (Desktop)**